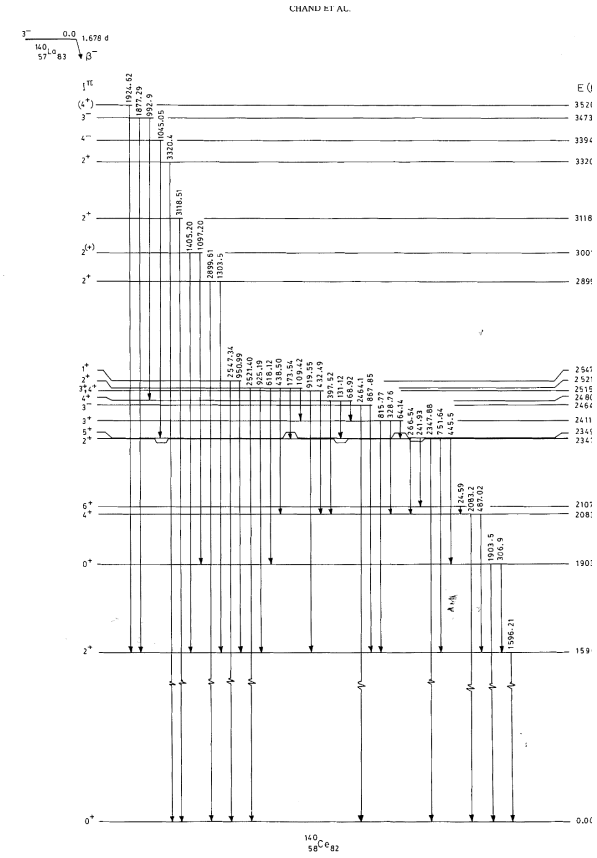
Four new transitions in ^{140}Ce

A compare-and-contrast of Chand 1991 vs. Mattera 2025

Overview

Mattera et al. (2025) added four γ-ray transitions to the decay scheme of ^{140}La $\rightarrow$ ^{140}Ce that were absent from Chand et al. (1991) and the current ENSDF evaluation. The intensities span two orders of magnitude below the Chand sensitivity floor.

E_γ (keV)	Transition	$J_i^\pi \rightarrow J_f^\pi$	I_γ (Mattera)	vs. Chand floor ($\sim$0.003)
166.1	2516 $\rightarrow$ 2350	$4^+ \rightarrow 5^+$	0.019(1)	**6$\times$ above**
644.5	2547 $\rightarrow$ 1903	$1^+ \rightarrow 0^+$	0.0023(2)	below
982.8	3395 $\rightarrow$ 2412	$(4^-) \rightarrow 3^+$	0.009(2)	$\sim$3$\times$ above
1009.7	3474 $\rightarrow$ 2464	$3^- \rightarrow 3^-$	0.008(1)	$\sim$3$\times$ above



(a) Chand 1991 decay scheme (their Fig. 4).

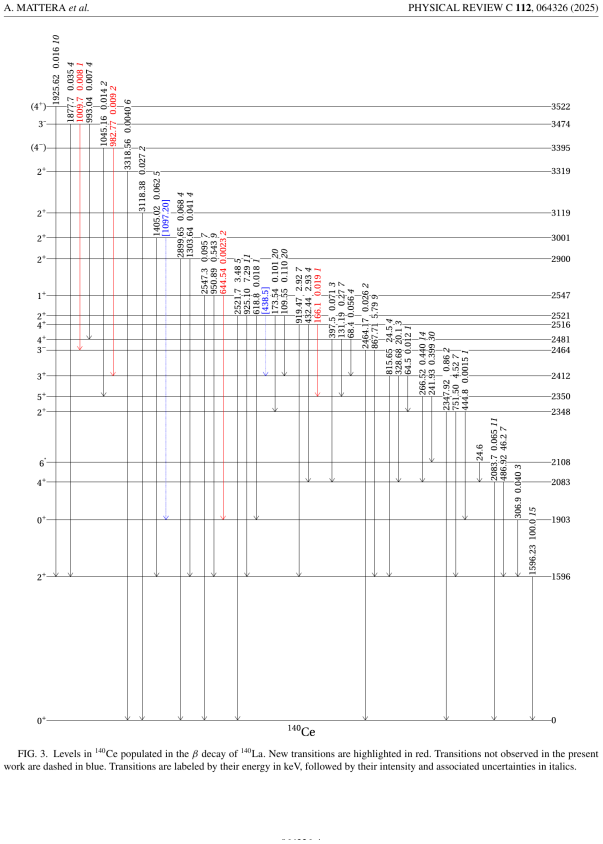


FIG. 3. Levels in ^{140}Ce populated in the β decay of ^{140}La. New transitions are highlighted in red. Transitions not observed in the present work are dashed in blue. Transitions are labeled by their energy in keV, followed by their intensity and associated uncertainties in italics.

(b) Mattera 2025 decay scheme (their Fig. 3). New transitions in red; disproved transitions in blue dashed.

Figure 1: Side-by-side decay schemes. Chand has 37 transitions; Mattera has 40 confirmed transitions, plus two Chand-era lines removed (438.5, 1097 keV).

1. The 166 keV transition: 2516 → 2350

Chand 1991

Table 3 of Chand: no entry at 166 keV. Nearby entries: 131.1 keV (I=0.49), 173.5 keV (I=0.13).

Chand level scheme: the 2516 keV level shows only two depopulating transitions — 432.49 keV (→ 2083) and 919.55 keV (→ 1596). **No 166 keV arrow.**

Chand Section 4.3: Chand explicitly discusses the 2516 keV level but only for its spin assignment (3^+ vs 4^+ ambiguity), not for additional decay channels. They had no reason to search for a 166 keV transition.

Mattera 2025

Mattera gate on both the 242 keV and 267 keV transitions that depopulate the 2350 keV level. If a 2516 → 2350 transition exists, it must be in coincidence with both. It is.

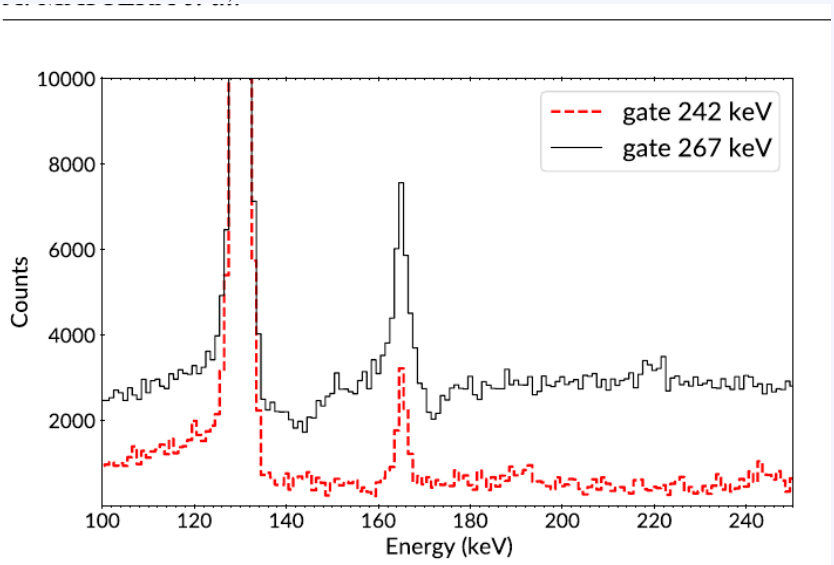


FIG. 4. A new transition at 166 keV is proposed that depopulates the level at 2516 keV, which was confirmed by gating on the two

A clean 166 keV peak appears in both gated spectra. The strong 131 keV peak at left is the known 2481 → 2350 transition feeding the 2350 level from above — a sanity check that the gate is working.

History: Hassan et al. 1986 had *proposed* this transition from coincidence with 1596 keV, but ENSDF rejected it for insufficient evidence. Mattera now confirms it.

Why Chand missed it

$I_\gamma = 0.019$ is 6× above Chand's nominal singles floor, so they had the raw sensitivity. What they lacked was **local resolvability**: 166 keV sits in the shoulder between the strong 131 and 173 keV peaks and on top of Compton continuum from every transition above 166 keV. In singles the peak is buried; in a coincidence gate that rejects uncorrelated continuum, it jumps out clearly. Chand didn't do a systematic gated search in this region because they weren't

suspecting a transition there.

2. The 645 keV transition: 2547 → 1903

Chand 1991

Table 3 of Chand: no entry at 645 keV. (Closest entries are 618.1 keV, I=0.039, and 751.6 keV, I=4.54.)

Chand level scheme: the 2547 keV (1^+) level shows only two depopulating transitions — 2547.34 keV (→ g.s.) and 950.99 keV (→ 1596). **No 645 keV arrow.**

Intensity context: at Chand's reported floor ($I \sim 3 \times 10^{-3}$), a peak at $I_\gamma = 0.0023$ is already at or below detection threshold, *even in a clean region.*

Mattera 2025

Mattera gate on the 307 keV transition that depopulates the 1903 keV (0^+) level. Any new transition feeding the 1903 level must be in coincidence with 307.

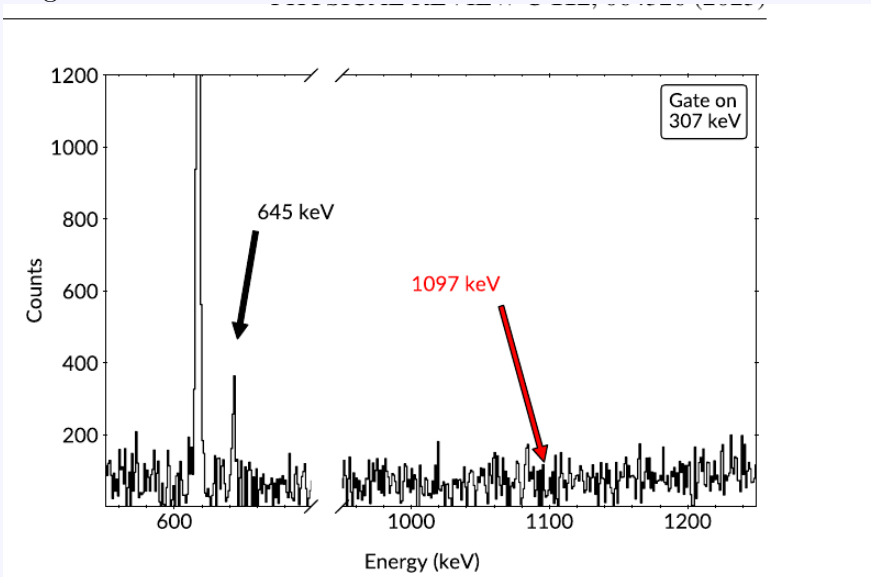


FIG. 6. Spectrum obtained with a gate on the 307 keV peak

Two things are shown on this plot: (i) a clean 645 keV peak is present (new transition *confirmed*); (ii) no peak at 1097 keV (Chand's proposed 3001 → 1903 transition, I=0.025, *disproved*; upper limit < 0.0002, 125× lower).

Why Chand missed it

Two compounding problems for Chand:

1. The 645 keV intensity ($I_\gamma = 0.0023$) is **at or below** Chand's statistical detection floor. This is the weakest of the four new transitions.
2. Confirmation requires gating on the **already-weak** 307 keV transition ($I_\gamma = 0.04$). Gating on a weak transition with a single HPGe pair gives you a gated spectrum with very few

counts — infeasible to look for 10^{-3} features.

The same figure also exposes a Chand-era error: the 1097 keV transition Chand reported at $I = 0.025(4)$ doesn't exist. Coincidence gating is simultaneously a discovery tool and a fake-peak killer.

3. The 983 keV transition: 3395 → 2412

Chand 1991

Table 3 of Chand: entry at 992.9 keV, I=0.014(5) — but this is the known 3474 → 2481 transition, *not* a 983 keV transition.

Chand level scheme: the 3395 keV (4^-) level shows only *one* depopulating transition — 1045.05 keV (→ 2350). **No 983 keV arrow.**

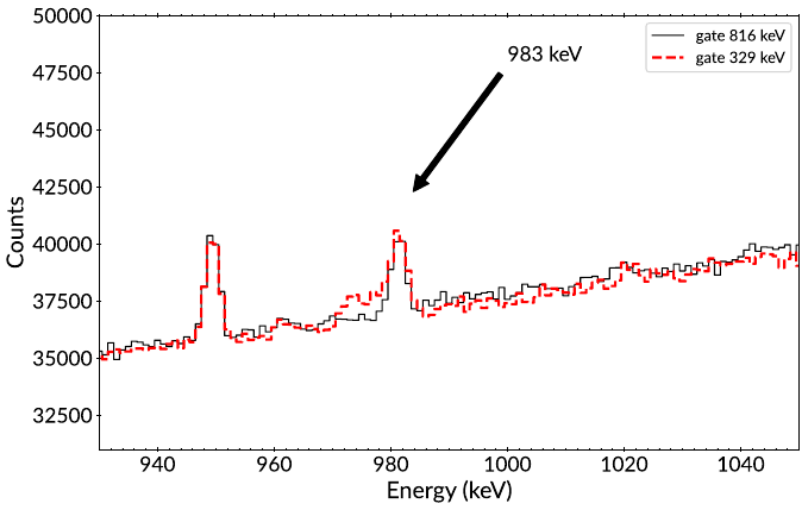
Concluding remarks (Chand, p. 100): *"The presence of the 1045.0 and 1924.6 keV transitions confirms the existence of the 3399.4 and 3520.8 keV levels, respectively."* Chand treated the high-lying levels as merely "confirmed" by their strongest depopulating transitions — not fully characterized.

Mattera 2025

Mattera gate on the 816 keV and 329 keV transitions, both of which depopulate the 2412 keV level. A new 3395 → 2412 transition must be in coincidence with both.

Appendix.

The strongest transition at 1596 keV is often reported with an intensity of 100, to which all other intensities are normalized. To convert these to absolute values, one can obtain a normalization factor by summing the total flux of transitions populating the ground state, of which the 1596-keV



A peak at 983 keV appears in both gates. The stronger peak at ∼950 keV is the well-known 951 keV transition (2547 → 1596), providing a reference on the same spectrum.

Why Chand missed it

Two reasons Chand missed this:

1. **Intensity:** $I_\gamma = 0.009$ sits near Chand's floor.
2. **Spectral crowding:** there is a *known* 993 keV transition (3474 → 2481, I=0.007) only 10 keV away. In a singles spectrum these two lines partially overlap and look like one broad

feature. Coincidence gating separates them cleanly because they come from different parent levels: gating on 816 or 329 (out of 2412) picks up 983 but rejects 993; gating on 868 (out of 2464) picks up the single-escape of 1596 but not 983.

Chand saw the 993 keV peak and attributed all of it to the 3474 → 2481 cascade. Some fraction of "their 993" may actually have been unresolved 983+993 contribution — a subtle systematic error that coincidence gating automatically corrects.

4. The 1010 keV transition: 3474 → 2464

Chand 1991

Table 3 of Chand: no entry at 1010 keV. Their entry at 1045.0 keV (I=0.026(15)) is the known 3395 → 2350 transition.

Chand level scheme: the 3474 keV (3^-) level shows only *two* depopulating transitions — 992.9 keV (→ 2481) and 1877.29 keV (→ 1596). **No 1010 keV arrow.**

Concluding remarks: same as for the 3395 keV level — Chand treated the 3474 keV level as merely confirmed by its strongest transitions.

Mattera 2025

Mattera gate on the 868 keV transition, which depopulates the 2464 keV (3^-) level. A new 3474 → 2464 transition must be in coincidence.

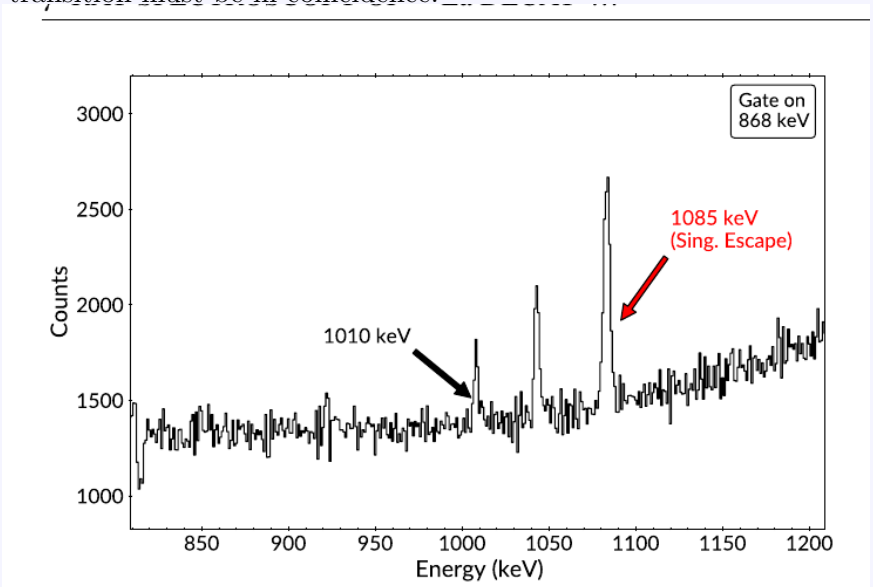


FIG. 8. A new 1010-keV transition is proposed depopulating the level at 3474 keV, confirmed by gating on the 868-keV transition

A clean 1010 keV peak is present. **Crucial detail:** the strong peak at 1085 keV is labeled as the **single-escape peak of the 1596 keV line** ($1596 - 511 = 1085$). This is a pure detector artifact — not a real transition — and its presence near 1010 keV is exactly what would have obscured the real line in a 1991 singles spectrum.

Why Chand missed it

The 1010 keV case is pedagogically the cleanest of the four because it illustrates **detector-physics contamination**, not just statistical limits:

- $I_\gamma = 0.008$ is at Chand's statistical floor.
- The 1085 keV single-escape peak from the very intense 1596 keV transition lives only 75 keV away. In singles, this escape peak is a *large* feature that creates a complicated local background structure.
- Gating on 868 keV isolates 1010 because the 1596 escape peak is not in coincidence with 868 — a physics-based separation that no amount of energy resolution in singles could achieve.

Moral: energy resolution and counting statistics weren't the limiting factors for this transition — the limit was *spectroscopic*, and only cascade correlations could break it.

Synthesis: the three mechanisms, visualized

Across the four transitions, Chand missed each one for one or more of three distinct reasons:

Transition	**(A)** Intensity below Chand floor?	**(B)** Buried in local continuum/shoulder?	**(C)** Contaminated by nearby peak/artifact?	Mattera tool
166 keV	No (6× above)	**Yes** (between 131, 173)	No	Gate on 242, 267
645 keV	**Yes** (below)	No	No	Gate on 307
983 keV	Near floor	No	**Yes** (993 keV)	Gate on 816, 329
1010 keV	Near floor	No	**Yes** (1085 SE)	Gate on 868

Answer to Q7

Three distinct mechanisms, one tool that solves all of them.

- **(A) Statistical limit.** The 645 keV transition is simply below what a two-HPGe-pair coincidence setup can accumulate at reasonable run times. Gammasphere's 1128-pair coincidence statistics reach deeper by ∼1000×.
- **(B) Local burial.** The 166 keV transition is statistically visible but sits in a region where Compton continuum and nearby strong peaks drown it out. Coincidence gating removes the continuum itself, not by improving the detector but by physically selecting correlated events.
- **(C) Spectral contamination.** The 983 and 1010 keV transitions overlap with nearby features (a known transition at 993 keV; a detector single-escape peak at 1085 keV). Cascade correlations separate these because the contaminant isn't fed by the same parent level as the signal.

Chand et al. (1991) ran a careful and well-instrumented measurement by 1991 standards. They had real coincidence capability (two HPGe detectors, 7 ns timing) but used it for **angular correlation of known cascades**, not systematic discovery. Mattera et al. (2025) run the same basic physics — HPGe coincidence spectroscopy, RADWARE analysis, angular correlations — but with 48 detectors producing 2.5×10^9 coincidence events to search. The methodology, not just the hardware, is what changed.